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REVIEW

A comprehensive review of deep learning techniques for image tampering detection

Piyush Kumar Singh[1] · Dheeraj Kumar[1] · Prabhat Ranjan[1]

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Abstract
Image tampering detection (ITD) represents a critical area of research that addresses the challenges posed by advanced digital alteration tools in evaluating the authenticity and integrity of visual media. This review conducts a thorough analysis of ITD techniques, categorizing them into two primary types: Active methods, which utilize embedded information for verification, and Passive methods, which analyze image characteristics without prior embedding. It highlights the transformative impact of deep learning architectures such as convolutional neural networks (CNNs), encoder-decoder models, and recurrent neural networks (RNNs) in enhancing both tampering localization and classification accuracy. Furthermore, this analysis examines the reliability of recent innovations, including hybrid models and feature extraction techniques, especially when applied to benchmark datasets such as CASIA and CoMoFoD. These datasets facilitate rigorous evaluation based on metrics such as precision, recall, and F1-score. Despite significant progress, challenges persist in developing algorithms that are resilient to sophisticated manipulation techniques, achieving real-time processing efficiencies, and enabling lightweight implementations suitable for edge devices. The review underscores the importance of interdisciplinary collaboration and the integration of explainable artificial intelligence in advancing digital image forensics across various applications, including journalism, forensic investigations, and ensuring the integrity of social media content.

Keywords Image tampering detection · Deep learning · Copy-move image · Spliced image

1 Introduction

In the contemporary digital landscape, visual representations are integral to the processes of communication, documentation, and entertainment. The widespread availability of digital devices and the advancement of editing software have significantly enhanced the ease and sophistication of image manipulation techniques. This situation has resulted in increasing apprehension regarding the authenticity and integrity of visual content, given that altered images may be employed to deceive, defraud, or disseminate misinformation (Korus 2017). Image tampering detection (ITD) represents a significant domain of inquiry focused on the advancement of methodologies and instruments designed to recognize and reveal modifications within digital images. Statistical analysis and Hand-crafted features are based upon the Traditional approaches, these are effective to an extent, but often struggle to keep pace with the increasingly complicated techniques used for tampering. Multimedia forensics (Verdoliva 2020), focused on validating the integrity of digital content, has garnered significant academic interest. Image tampering, referred to as image forgery, manipulation, or forensics, is a research domain focused on the analysis of manipulated images. This field addresses complex tasks, including the localization of tampered regions, validation of image integrity, and identification of the source or provenance of altered images (Park and Choeh 2018; Park et al. 2016; Rao et al. 2020). Deep learning (DL), which provides strong feature extraction and pattern recognition tools, has transformed several areas of computer vision in recent years (Parker 2010; Chen et al. 2024). DL-based

✉ Dheeraj Kumar
dheerajkumar@cusb.ac.in

Piyush Kumar Singh
piyush@cusb.ac.in

Prabhat Ranjan
prabhatranjan@cub.ac.in

[1] Department of Computer Science, Central University of South Bihar, NH-120, Gaya Panchanpur Road, Gaya, Bihar 824236, India

approaches have demonstrated exceptional efficacy in tasks such as image classification, object detection, and semantic segmentation, making them auspicious avenues for advancing ITD.

The terms “image manipulation", “forgery", and “tampering" refer to various illegal changes or alterations made to digital images. Every word describes a unique manipulation involving various goals and methods. Below is a definition for each word:

- *Image Manipulation:* Intentionally changing digital photographs to provide a desired effect or visual impact is known as image manipulation (Park and Choeh 2018). To improve the image’s visual quality or provide a particular artistic effect, this technique entails adjusting some characteristics, including color, contrast, brightness, and sharpness. The most frequent way to change images is to utilize tools for image manipulation, such as Adobe Photoshop or GIMP. Any intentional alteration intended to trick or mislead viewers can also be considered image manipulation within the realms of computer forensics and integrity verification (Park et al. 2016; Rao et al. 2020).
- *Image Forgery:* The creation or modification of an image with the specific goal of misleading or deceiving viewers is referred to as image forgery. The method usually entails fabricating false or phony photographs that appear legitimate (Park et al. 2016; Park and Choeh 2018). Forgery can take many different forms, such as creating completely fake photos, copying or removing parts of one image, or altering particular regions of an image. The creation of a false narrative, the manipulation of evidence, or the distortion of information are frequently the main goals of image forgery. In fields like forensic research, journalism, and legal processes, identifying image forgeries is an essential task.

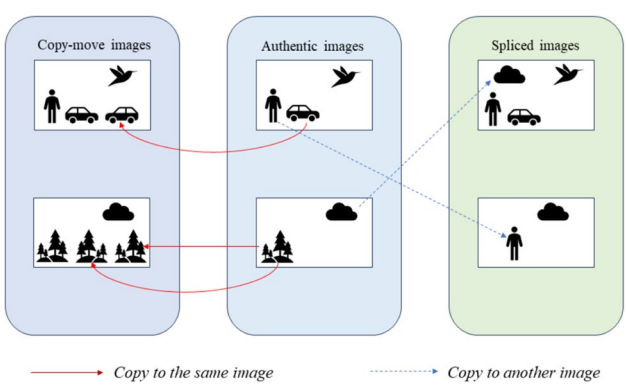

Fig. 1 The representation shows the arrangement of two copy-move photos on the left and two spliced photos on the right, which are derived from two original images positioned in the center

- *Image Tampering:* “Image tampering” describes the illegal modification or restructuring of digital images without the permission or awareness of the owner or the intended viewers. Potential modifications include manipulating information to deceive users by relocating or omitting elements, modifying content, or altering specific areas of an image. Tampering can manifest in several ways, such as altering private or malicious images, fabricating false evidence, or disseminating deceptive information (Park and Choeh 2018; Park et al. 2016). The preservation of the reliability and legitimacy of digital photographs relies on the capacity to identify and detect image alteration (Rao et al. 2020; Sharma et al. 2024; Asghar et al. 2017). Figure 1 illustrates the fundamental differences between the two techniques. “Copy-move” involves duplicating a portion of an image and pasting it onto the same image, whereas “Splicing” involves copying a region from one image and pasting it into a completely different one.

In all of these scenarios, digital forensic techniques, algorithms for image processing, and methodologies for machine learning are frequently utilized for identifying and examining discrepancies, abnormalities, or remnants resulting from manipulation or tampering (Singh and Kumar 2024; Mahto et al. 2024). The methods used involve scrutinizing analysis at the pixel level, performing information analysis, employing image transforming, and implementing watermarking to detect signs of alteration or manipulation of the photos (Mahto et al. 2024). The act of altering, faking, and tampering with images generally highlights how crucial it is to maintain the veracity, honesty, and dependability of electronic pictures, especially in domains such as media coverage, criminal justice, and court cases, where accurate proof and presentation are crucial (Park and Choeh 2018; Park et al. 2016; Rao et al. 2020; Sharma et al. 2024; Asghar et al. 2017).

While several surveys have reviewed image tampering detection, most focus either on traditional approaches or provide only high-level overviews of deep learning techniques without critically assessing their limitations. Existing works often lack a consolidated comparison of models, datasets, and evaluation metrics, making it difficult for new researchers to quickly identify strengths and weaknesses across approaches. Moreover, the role of explainable AI and emerging lightweight architectures for real-time or edge-based applications remains underexplored in prior reviews. This study addresses these gaps by providing a comprehensive and critical synthesis of deep learning strategies for image tampering detection, supported by detailed comparison tables, dataset analyses, and a discussion of future directions that emphasize explainability and robustness.

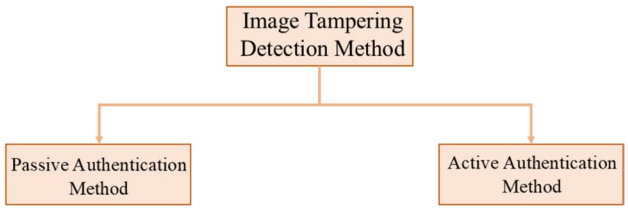

Fig. 2 Taxonomy of image tampering detection methods, categorized into active and passive approaches

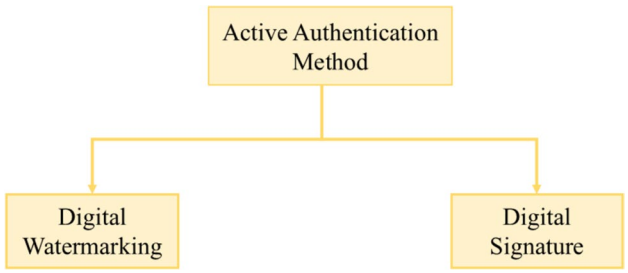

Fig. 3 Schematic of active authentication methods, including digital watermarking and digital signatures

The remainder of this paper is organized as follows. Section 2 provides a detailed analysis of strategies employed for image tampering detection. Section 3 presents a literature review and DL-based ITD methodologies along with the experimental outcomes of the reviewed approaches. Section 4 discusses the deep learning backbone architectures relevant to ITD. Section 5 introduces widely used benchmark datasets for ITD challenges, while Section 6 outlines the evaluation standards and metrics applied to assess ITD methodologies. Finally, Section 7 highlights prospective future research directions and concludes the study.

2 Image tampering detection methods

According to Fig. 2, the classification of detecting digital image (Castleman 1979) tampering can be separated into two different categories: active techniques, which are not blind, and passive techniques, which are blind (Sharma et al. 2024; Asghar et al. 2017).

2.1 Active authentication method

This method entails inserting information into an image while it is being created. It can thereafter be employed to ascertain the image's origin or identify any possible modifications. Digital watermarking and digital signatures (Ch et al. 2024) are the two categories into which active authentication techniques fall. Digital watermarking is a process that embeds a bit of pattern within a digital image (Castleman 1979) while it is being captured or processed. Digital signatures (Castleman 1979) embed specific data taken from the same image into the image itself. This method has the benefit of having less processing overhead, and fraud detection can be easily accomplished if the original image's metadata is provided. The disadvantage of these approaches is that they need to be embedded into photos during a moment of acquisition utilizing specialized technology, and confirmation requires prior familiarity with the image (Sharma et al. 2024; Singh and Chadha 2013). Figure 3 illustrates the two primary categories of active authentication.

2.2 Passive authentication method

There is no need to include extra data when creating a picture while using passive authentication (Wang et al. 2017). Genuine images have inherent patterns that are constant from imaging devices or processing methods. These patterns are disturbed by any modification. Also called "blind authentication", it verifies integrity only using picture features and does not require the original. It lacks real-time data, such as fingerprints or watermarks, and therefore is inert. In true photos, manipulation affects the statistical patterns, typically undetectable to the human eye. This results in the identification of inconsistencies through the use of basic image processing techniques (Birajdar and Mankar 2013; Kaur et al. 2024). Below is a brief explanation of each classification in Fig. 4.

1. *Pixel-Based Tampering Detection:* Pixel-based approaches (Qureshi and Deriche 2015) include analyzing correlations at the smallest level, which appear in the spatially or altered region throughout alteration,

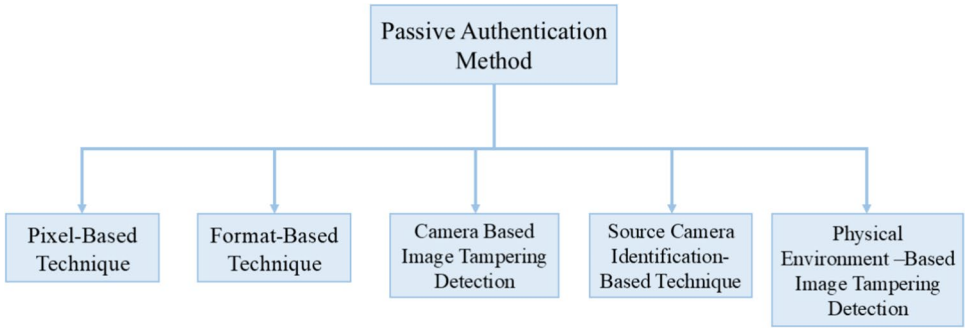

Fig. 4 Schematic of passive authentication methods, which detect tampering based on intrinsic image features without prior embedding

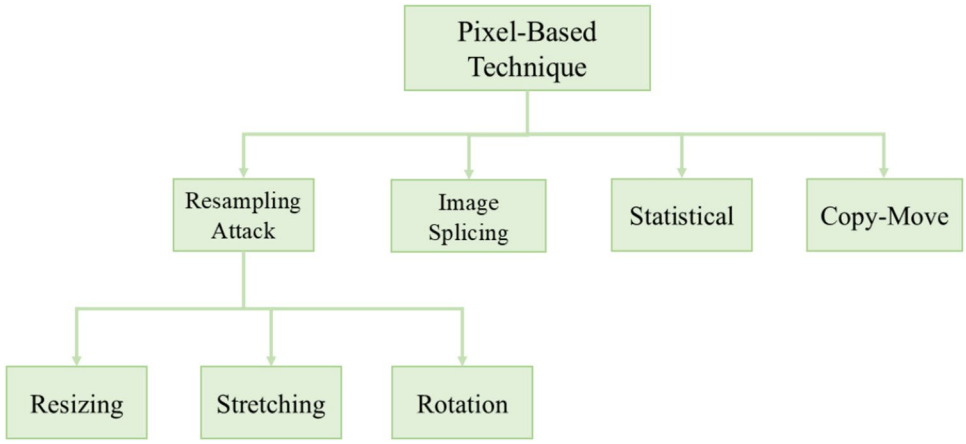

Fig. 5 Categories of pixel-based tampering detection: erase-fill, copy-move, splicing, and resampling

Fig. 6 Example of copy-move tampering: **a** authentic image, **b** tampered copy-move image, and **c** corresponding ground truth mask

and detecting statistical variations at the pixel level. Pixel-based tampering identification techniques have been categorized into four distinct groups (Ansari et al. 2014) as seen in Fig. 5. Figure 6 presents a visual breakdown of a copy-move forgery. It displays the authentic image (a), the tampered version where an object has been duplicated (b), and the corresponding ground truth mask (c) used to localize the forged pixels.

- *Erase-fill:* The erase-fill was accomplished through the use of brushes, blurring, or modest patch replication to obscure sections of the original image (Ansari et al. 2014).
- *Copy-Move:* The copy-move was executed by duplicating and pasting a specific section or object onto the same image, with or without scaling and rotation modifications (Ansari et al. 2014).
- *Splicing of image:* In order to alter the uniqueness of an image, splicing was accomplished by copying and pasting a section of one image over another (Ansari et al. 2014).
- *Resampling :* One person may need to be stretched or resized to meet the other individual's relative heights when creating a composite of two people. Thus, the original image must be resampled into a new sampling lattice in this procedure (Qureshi and Deriche 2015; Ansari et al. 2014).

2. *Format-Based Tampering Detection:* The main component of this method is the JPEG format. To identify image forgeries, it makes use of the quantifiable relationships that arise from specific lossy compression. These techniques fall into three categories: double JPEG, JPEG blocking, and JPEG quantization (Bourouis et al. 2020). When a picture is compressed, it is quite difficult to distinguish between a counterfeit and an original. Nonetheless, out of the compressed photos that are available, forged images can be identified using format-based approaches (Ding et al. 2020. Figure 7 outlines the taxonomy of format-based techniques.

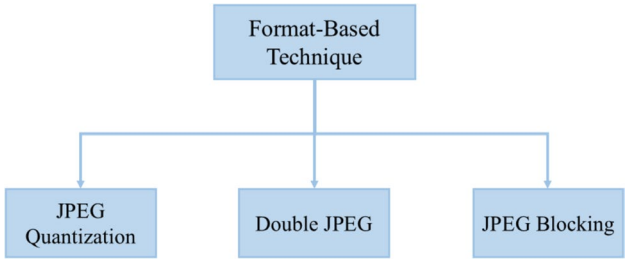

Fig. 7 Overview of format-based tampering detection using JPEG compression artifacts

3. *Camera-Based Tampering Detection:* This procedure identifies modifications introduced during the formation of an image by scrutinizing the variances introduced by diverse phases of the imaging procedure. This includes many approaches such as quantization, color correlation, gamma correction, white balance, filtration, and JPEG compression (Bondi et al. 2017). Below is a brief explanation of each classification in Fig. 8.

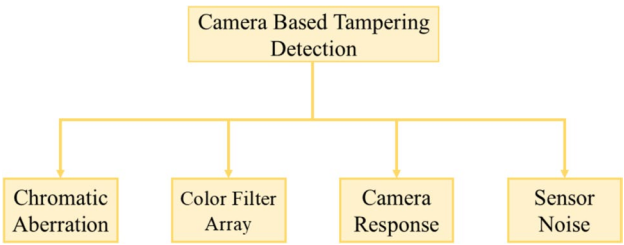

Fig. 8 Camera-based tampering detection framework exploiting sensor and imaging pipeline inconsistencies

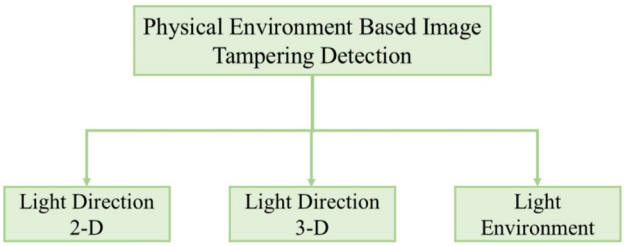

Fig. 9 Physical environment-based tampering detection, focusing on illumination inconsistencies across manipulated regions

4. *Source Camera Identification-Based Technique:* The camera's image-capturing parameters are employed to detect tampering (Zheng et al. 2024). This approach involves analyzing several factors, such as sensing the background noise and lens aberrations (Nwokeji et al. 2024).
5. *Physical Environment-Based Image Tampering Detection:* In photography that captures the natural world or depicts a realistic image, one is likely to encounter a range of lighting conditions. These fluctuations in the brightness and contrast of the original image can function as clues to detect any form of manipulation intended to produce an altered image. A highly effective means of identifying image forgery lies in carefully examining the variations in illumination within the manipulated picture and the initial version (Chennamma and Madhushree 2023). Consider a modified image of two people walking along a beach. Their original images were taken in different settings. One can determine the image's authenticity or fabrication by evaluating its illumination (Deb et al. 2024). Figure 9 illustrates the classification of these methods, distinguishing between 2-D Light Direction, 3-D Light Direction, and Light Environment analysis to detect illumination anomalies.
6. *Geometry-Based Image Tampering Detection:* The lens of a camera concentrates light coming from an object during image capture. The camera sensor chip subsequently receives this light and decomposes it into several separate pixels. Thereafter, the sensor measures the hue and luminance of every pixel and documents these data as quantitative values. Every measurement is made with the primary point or center of the camera, which appears on the visual plane (Kumari and Garg 2024). Figure 10 shows the framework for geometry-based detection, which relies on the camera's optical physics.

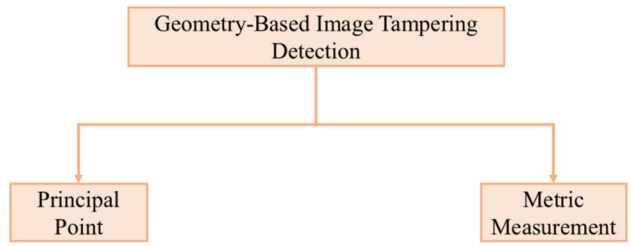

Fig. 10 Geometry-based tampering detection, identifying manipulations through inconsistencies in intrinsic camera parameters

A pure image is produced when the camera's focal position aligns with the image's center on the image plane. However, object movement, translation shifts, and picture segment merging to form a composite image will affect the principal point's position in the image. Tampering can be detected by varying internal metric characteristics, including skew, aspect ratio, focal length, and primary point in an untouched image. Tampering detection methods can be developed using projective geometry concepts (Bourouis et al. 2020). Tampering detection is multi-dimensional. Copied regions in modified photographs can often be identified without the original. In copy-move tampering, merged pictures are created by making a copy of a picture and adding a portion that is identical to the original (Ghai et al. 2024).

In this Section, we provide a brief overview of the ITD methods and their types. By comparing active and passive techniques, this Section concludes by emphasizing the diverse approaches available for detecting tampered regions, setting the foundation for subsequent discussions on deep learning advancements.

3 Literature review

A comprehensive review of existing literature is essential to position recent developments in image tampering detection (ITD) within the broader research landscape. Early studies primarily relied on handcrafted features and statistical analysis to identify inconsistencies caused by tampering, such as illumination irregularities or copy-move artifacts. While these approaches provided valuable insights, their performance was often limited by a lack of generalization and sensitivity to post-processing operations. Early surveys in multimedia forensics primarily emphasized active methods such as watermarking and digital signatures, which provided reliable detection but required embedding information at acquisition time (Ch et al. 2024; Singh and

Chadha 2013). These approaches were limited in scalability, as they relied on controlled image capture and prior access to original data.

Subsequent research shifted to passive (blind) techniques, which analyze image characteristics such as noise patterns, illumination inconsistencies, and compression artifacts (Wang et al. 2017; Qureshi and Deriche 2015). Surveys by citekorus2017digital and Verdoliva (2020) provide taxonomies of these methods, highlighting their utility in detecting splicing and copy-move forgeries. While these methods improved general applicability, their dependence on handcrafted features restricted robustness against increasingly sophisticated manipulations.

The introduction of deep learning marked a paradigm shift in ITD. CNNs, encoder-decoder networks, and RNNs enabled automatic feature extraction and end-to-end learning for forgery localization and classification (Alzahrani 2024; Schetinger et al. 2017). Verdoliva (2020) and recent surveys stress that DL models outperform traditional techniques across benchmarks such as CASIA and CoMoFoD, particularly for splicing and copy-move tasks. However, they also caution that overfitting to specific datasets and the high computational burden limit real-world deployment.

More recent comparative studies (Aribilola et al. 2024; Choudhary et al. 2024; Zhang et al. 2023; Fahmi and Sari 2021; Manjunatha and Patil 2021; Chakraborty et al. 2024; Dong et al. 2013; Walia and Kumar 2019; Warif et al. 2022; Zhang et al. 2024; Tyagi and Yadav 2023; Sharma et al. 2023; Sadanand et al. 2024; Vijayalakshmi K et al. 2024; Alencar et al. 2024) have evaluated the strengths of different DL architectures, ranging from lightweight dual-branch CNNs to complex hybrid models combining segmentation and classification. These works reveal that while CNN-ELA methods achieve the highest accuracies (> 99%) in constrained datasets, hybrid models such as VGG16-UNET provide superior localization. DenseNet and ResNet variants mitigate vanishing gradient issues but introduce higher computational costs. Collectively, these findings underscore the trade-off between accuracy, interpretability, and efficiency. Despite these advances, there remain important research gaps. Existing surveys often lack critical evaluations of limitations, particularly in terms of cross-dataset generalization, adversarial robustness, and interpretability. Few works systematically integrate evaluation metrics and dataset characteristics into their analysis, which are vital for practitioners seeking reproducibility.

3.1 Review of DL-based strategies for image tampering detection

In contrast, modern deep learning (DL) methods have emerged as a dominant paradigm, offering improved feature representation and robustness against diverse manipulations. Architectures such as convolutional neural networks (CNNs), autoencoders, and densely connected networks (DenseNet) have achieved state-of-the-art performance across widely used datasets. This section reviews representative DL-based and classical approaches, summarizing their design, datasets, evaluation metrics, and contributions to advancing ITD research.

Chakraborty et al. (2024) propose a dual-branch CNN that leverages Error Level Analysis (ELA) and Spatial Rich Model-based noise residuals to achieve 98.55% accuracy on the CASIA v2.0 dataset (Dong et al. 2013). This lightweight model highlights the advantage of combining handcrafted features with deep learning for efficient and robust detection. Sadanand et al. (2024) focus on integrating CNNs with ELA, achieving an even higher accuracy of 99.7% across multiple datasets (CASIA v1.0, v2.0, MICC) (Dong et al. 2013; Amerini et al. 2011). This work emphasizes the utility of ELA in detecting compression artifacts and pixel inconsistencies. Alencar et al. (2024) introduce a multi-stream neural network architecture that uses passive methods such as ELA and DWT for feature extraction. Using a novel curated dataset with multiple tampering types, the approach achieved 89.59% accuracy while addressing dataset bias challenges. Saxena et al. (2024) refine CNN applications by incorporating separable convolution layers into a custom architecture, optimizing computational efficiency and achieving a 96% F1-score for detecting sophisticated manipulations. Vijayalakshmi K et al. (2024) propose a convolutional autoencoder combined with ELA for copy-paste forgery detection, demonstrating 99.2% precision on the MICC-F220 dataset. Alzahrani (2024) improves DenseNet-based CNNs for forgery detection, addressing vanishing gradient and over-complexity issues. Using CASIA v2.0, the model achieved 92.32% accuracy, outperforming traditional methods in precision and efficiency. Choudhary et al. (2024) design a hybrid VGG16-UNET model for forgery localization, integrating transfer learning with segmentation. Achieving 91.7% accuracy on CASIA v2.0, it provides effective localization of tampered regions, though its scope is largely limited to splicing detection. Rafique et al. (2023) develop a hybrid framework combining ELA with pre-trained CNNs (GoogLeNet, ResNet18, SqueezeNet) and classical classifiers (KNN, SVM). The approach achieved 89.5% accuracy on a deepfake dataset, showcasing the robustness of ELA preprocessing but limited scalability for large-scale applications. Kaur et al. (2023) introduce a copy-move forgery detection method combining CLAHE and CNN for improved preprocessing and feature extraction. Demonstrating high accuracy across multiple datasets, it withstands geometric attacks such as rotation and compression. Vaishali and Neetu (2024) employ a ResNet-101

architecture with cyclical learning rates (CLR) to address vanishing gradient issues. Achieving 97.75% accuracy on multiple datasets, their model reduces training epochs and enhances efficiency for real-time deployment.

Beyond image-based studies, researchers have also examined the perceptual aspects of manipulation. Schetinger et al. (2017) investigated human ability to detect erase-fill, copy-move, and cut-paste tampering, reporting varying detection rates (38.5%−59.4%). Traditional feature-based methods, such as texture descriptors, Local Binary Patterns (LBP), and Median Robust Extended Local Binary Patterns (MRELBP), have also been applied to splicing detection, demonstrating improvements in boundary localization (Manu and Mehtre 2019).

Niu et al. (2021) proposed a keypoint-based hierarchical matching approach using discriminative moment invariants for efficient copy-move detection, while Yuan et al. (2021) employed Gauss-Jordan elimination-based watermark generation for tampering localization. da Costa et al. (2020) explored anomaly detection techniques, whereas Bonomi et al. (2021) applied Local Derivative Patterns on Three Orthogonal Planes (LDP-TOP) for detecting spoofing in facial images. Video tampering detection has also gained attention. Fadl et al. (2021a) used spatiotemporal CNN features with fusion techniques to detect inter-frame forgeries, including frame deletion, insertion, and duplication. Similarly, another study (Fadl et al. 2021b) combined 2D-CNNs with RBF-MSVMs for classifying video manipulations under compression and noise. More recently, watermarking-based methods have been explored for both images and videos. Sisaudia and Vishwakarma (2024) proposed a block-level watermarking scheme using LBP for tamper detection and recovery, while Padma Priya R and Krishna (2021) developed a chain-structured watermarking algorithm using SVD and IWT for surveillance video protection. Similarly, SecureCam Aribilola et al. (2023) integrated unsupervised learning with encryption for video forgery detection in mobile devices. Aribilola et al. (2024) introduced TampDetect, an HSV-based lightweight model for detecting encrypted video tampering, achieving 100% accuracy on Intel NUC devices.

Table 1 provides a consolidated comparison of recent deep learning-based strategies for ITD. While CNN variants, autoencoders, and DenseNet models achieve strong results on datasets such as CASIA and MICC, many face challenges in cross-dataset generalization and computational scalability. Hybrid architectures such as VGG16-UNET improve localization, and CLAHE-CNN models show robustness to geometric transformations. However, most methods remain specialized for copy-move forgeries, with limited coverage of splicing and emerging manipulations such as deepfakes.

Tables 1 and 2 together provide a holistic comparison of tampering detection techniques. Table 1 focuses on DL-based strategies with high accuracy and advanced feature representation, while Table 2 highlights classical and hybrid approaches such as watermarking, statistical modeling, and traditional feature extraction. This comparison shows how ITD research has changed over time, moving from interpretable but lightweight approaches with scalability issues to data-driven deep models that are highly accurate but computationally intensive. Several restrictions still exist in spite of these advancements. Concerns regarding real-world generalization are raised by the fact that many DL-based ITD techniques only show good performance on particular test datasets. Deployment in mobile and surveillance situations is limited by high processing complexity and dependence on sizable annotated datasets.

4 Overview of deep neural network architectures

Deep Learning (DL) has been the primary focus of research in multiple disciplines during the past decade, as evidenced by references (Vo et al. 2021a) and Vo et al. (2021b). Over the last five decades, the application of DL techniques has surpassed traditional physics-based methods in tackling Image Tampering Detection (ITD) issues, as evidenced by references (Kaur et al. 2024; Walia and Kumar 2019; Warif et al. 2022; Zhang et al. 2024) and Tyagi and Yadav (2023). The schematic representation of the generic deep learning-inspired pipeline for detecting image tampering is depicted in Fig. 11, showcasing that the complete input image or segmented patches are fed into an advanced Deep Neural Network (DNN) for the extraction of features. The two unique methodologies are known as the image-wise and patch-wise approaches. In patch-wise techniques, the segmented patches may be either concurrent or non-concurrent, each presenting distinct advantages. The output of this network is a binary picture that accurately identifies the altered areas in the input image.

Given that both image segmentation and image tampering detection involve pixel-based classifications, deep learning networks employed for one may also apply to the other (Sharma et al. 2023; Pham and Park 2023; Raveendra and Nagireddy 2022; Zanardelli et al. 2023). The two issues are closely related. Current advanced deep learning methodologies have achieved significant success in segmenting pictures into r regions with l labeled classes, where $r < l$ (Pham and Park 2023; Diakogiannis et al. 2020; Mo et al. 2022) for semantic segmentation of pictures. When $r = l = 2$, picture tampering detection is seen as a traditional form of image segmentation, wherein foreground objects and the

Table 1 Comparison of deep learning-based strategies for image tampering detection (ITD), summarizing the employed architectures, benchmark datasets, reported performance, key strengths, and identified limitations

Author and references	DL techniques used	Datasets	Accuracy / F1	Strength	Limitations
Chakraborty et al. (2024)	Dual-branch CNN with ELA and SRM	CASIA v2.0	98.55	Combines handcrafted features with CNN for efficient tampering detection. Lightweight model suitable for low-resource settings	Evaluated only on the CASIA dataset; lacks validation on diverse or large-scale datasets
Alencar et al. (2024)	Multi-stream CNN with ELA and DWT	Curated dataset (4 public datasets)	89.59	Improves feature extraction; demonstrates robustness with real-world manipulations	Does not provide detailed localization of tampered regions
Sadanand et al. (2024)	CNN with ELA	CASIA v1.0, CASIA v2.0, MICC	99.7	Demonstrates high-resolution tampering detection, effective for manually altered photos	Performance is highly dataset-dependent; it lacks cross-dataset validation and real-world scenarios
Saxena et al. (2024)	Custom CNN with separable convolutions	Defacto, COCO	F1 score: 96	Optimized for diverse manipulations; computationally efficient with strong validation	Dataset diversity and cross-validation with real-world images are not emphasized
Vijayalakshmi K et al. (2024)	Autoencoder + ELA	MICC-F220	99.2	Achieves high CPF detection; preprocessing reduces overfitting	Focused on copy-paste forgeries only; generalization across datasets is limited
Alzahrani (2024)	DenseNet-based CNN	CASIA V2.0	92.32	Addresses gradient vanishing; performs well on CASIA	Computationally expensive for large-scale datasets; lower accuracy than hybrid/custom models
Choudhary et al. (2024)	VGG16 + UNET + ELA	CASIA V2.0	91.7	Combines ELA and UNET for localization and classification; effective transfer learning	Focuses only on splicing; lacks evaluation across other forgery types
Rafique et al. (2023)	GoogLeNet, ResNet18, SqueezeNet + ELA	Yonsei Deep Fake Dataset	89.5	Robust against pixel-level manipulations; integrates CNNs with classical classifiers	Moderate accuracy; scalability for real-world deployment is limited
Kaur et al. (2023)	CLAHE + CNN	MICC-F220, MICC-F600, GRIP, IMD	94	Resistant to geometric attacks (rotation, compression); robust copy-move detection	High computational demands; limited focus on splicing and complex forgeries
Vaishali and Neetu (2024)	ResNet-101 with cyclical learning rate	MICC-F220, MICC-F600, MICC-F2000, CoMoFoD	97.75 (CoMoFoD)	Addresses vanishing gradients; efficient training with fewer epochs	Computationally expensive; limited testing on splicing/non-CMF forgeries

background are categorized into two distinct classes. Similarly, methodologies utilizing object detection, including Mask R-CNN and convolutional neural networks that are region-based (R-CNN), have been extensively employed in tampering detection. Deep learning (DL) approaches, including RNN and LSTM, are frequently employed in the fields of speech recognition (Pham and Park 2023; Doegar et al. 2019) and language modeling.

1. *Convolutional neural network:* CNN constitutes a significant category of DL architectures (Khan et al. 2020), especially created for the analysis of grid-structured data, including images. A standard convolutional neural network consists of three main types of layers: convolutional layers, pooling layers, and fully connected layers (O'Shea 2015). As demonstrated in Fig. 11, it is common for each convolutional layer to be succeeded by a pooling layer. This configuration serves to decrease the physical dimension of the data, consequently minimizing the computational complexity of

Table 2 Comparative analysis of existing tampering detection techniques beyond deep learning approaches, including classical, watermarking, and hybrid methods

References	Mode of tampering	Accuracy	Strength	Limitations
Aribilola et al. (2024)	Pixel injection, pixel modification	100%	Lightweight architecture (TampDetect) enables deployment on Intel NUC and IoT devices, allowing early detection before decryption	Limited assessment metrics and weak robustness against complex attacks; HSV models may fail under illumination changes
Sisaudia and Vishwakarma (2024)	Copy-move attack, cropping, text insertion, histogram equalization, blurring	98%	Proposes a real-time tamper detection and recovery method with quick reconstruction of tampered regions	Restricted to grayscale images; reconstruction quality may be insufficient for high-security applications
Lee et al. (2022)	Copy-move forgery	90.5%	Rotation-invariant wavelet features with CNN show robustness across datasets	Wavelet transform + CNN increases computational overhead; limited to copy-move forgeries
Zhang et al. (2023)	Geometric attacks, noise-based attacks, filtering, compression	99%	SIFT precisely localizes tampered areas, resists compression/rotation, and integrates reversible watermarking	Computationally intensive feature extraction; lacks validation on diverse datasets
Fadl et al. (2021b)	Frame deletion, insertion, duplication (video)	98%	Detects video forgery using spatiotemporal analysis and RBF-MSVM, resistant to noise, compression, and blurring	Does not test under environmental variations; requires tuning for large-scale surveillance datasets
Padma Priya R and Krishna (2021)	Image addition	97%	Lightweight blockchain + watermarking method reduces storage and computational requirements	Heavy reliance on watermarking; limited scalability testing compared to non-watermark methods
Fahmi and Sari (2021)	Cloning (copy-move), splicing, resampling, statistical attacks	80%	Compares CNN architectures and identifies challenges in passive forgery detection	Only considers copy-move; lacks augmentation or regularization for generalization
Manjunatha and Patil (2021)	Splicing, compression, rotation, resampling, copy-move	97.7%	Evaluates multiple datasets; highlights the role of pixel correlation and feature extraction	Lacks novel contributions or experimental validation; limited use of advanced deep learning methods

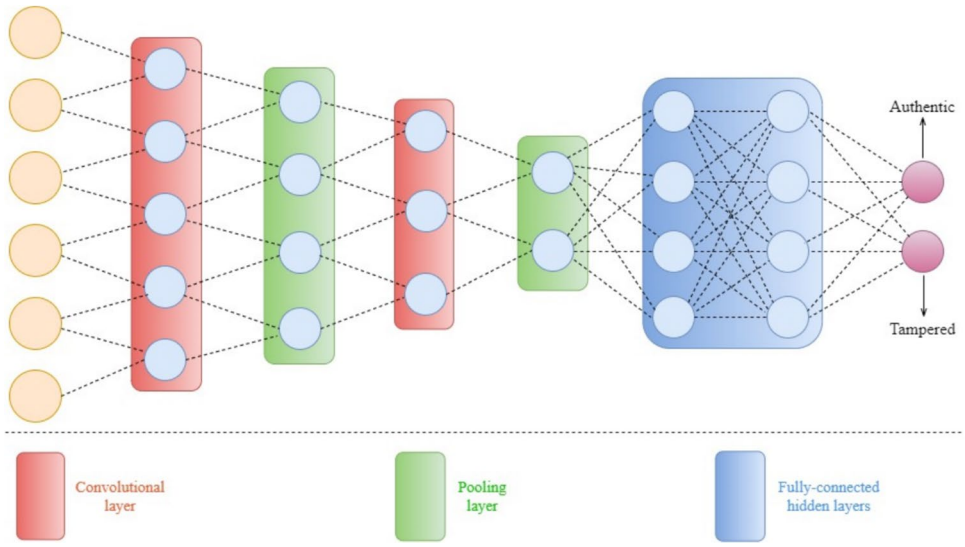

Fig. 11 A straightforward example employed a CNN architecture used to solve ITD issues. This illustration was reproduced from Neutelings (2021)

the following layers while maintaining critical features. In an entirely connected layer, each neuron forms connections with all neurons in both the previous and following layers. The integration of layers that are completely connected is essential in the decision-making frameworks of CNNs; however, it significantly raises the parameter count, which in turn amplifies the computational demands.

2. *Encoder-decoder and auto-encoder networks:* The encoder-decoder (Pham and Park 2023) constitutes a unique class of algorithms adept at obtaining efficient representations of input data without the requirement for labeled information (Chen and Guo 2023). This classification includes artificial neural networks specifically designed for unsupervised learning. An automatic decoder's fundamental principle is the capacity to compress and accurately represent input data without explicit labels. This is achieved through a bifurcated architecture comprising an encoder and a decoder. The input data must be transformed into a lower-dimensional representation by the encoder, commonly referred to as "latent space" or "encoding". Utilizing this representation, the decoder reconstructs the original input. To enable the network to discern significant patterns within the data, the iterative process of encoding and decoding plays a critical role in explaining essential features, Fig. 12. An Auto-encoder represents a unique variant of an encoder-decoder framework defined by a solitary hidden layer, wherein it seeks to regenerate the original inputs from the decoded representations. Auto-encoders are commonly employed to address many challenges, including but not limited to data denoising, classification, clustering, and the identification of anomalies (Bank et al. 2023).
3. *LSTM:* LSTM represents a specific category of recurrent neural networks (RNNs) that has been developed to address the vanishing gradient issue commonly encountered in conventional RNNs (Yu et al. 2019). It introduces gates to control the flow of information, allowing it to remember long-term dependencies. The cell state is the key concept in LSTM. It facilitates information flow across the chain by functioning as a conveyor belt. Information can be added or deleted from the cell state using gates. Hochreiter and Schmidhuber (1997),Staudemeyer and Morris (2019), Sherstinsky (2020). Fig. 13 shows an example of an LSTM architecture (https://blog.mlreview.com/understanding-lstm-and-its-diagrams-37e2f46f1714) for ITD problems. At each time step t, the LSTM takes the input vector x_t, the previous hidden state h_{t-1}, and the previous cell state c_{t-1}. The following computations are performed (Hochreiter and Schmidhuber 1997; Staudemeyer and Morris 2019; Sherstinsky 2020).
Forget Gate: The forget gate controls which information to discard from the previous cell state.

$$\mathrm{f}_t = \sigma\left(\mathrm{W}_f \cdot [\mathrm{h}_{t-1}, \mathrm{x}_t] + \mathrm{b}_f\right), \tag{1}$$

where σ is the sigmoid activation function, W_f is the weight matrix, and b_f is the bias vector.
Input Gate: The input gate decides which new information to add to the cell state.

$$\mathrm{i}_t = \sigma\left(\mathrm{W}_i \cdot [\mathrm{h}_{t-1}, \mathrm{x}_t] + \mathrm{b}_i\right), \tag{2}$$

$$\tilde{\mathrm{c}}_t = \tanh\left(\mathrm{W}_c \cdot [\mathrm{h}_{t-1}, \mathrm{x}_t] + \mathrm{b}_c\right), \tag{3}$$

where i_t is the activation of the input gate, and $\tilde{\mathrm{c}}_t$ represents the candidate cell state.
Cell State Update: The cell state is updated by combining the forget gate and the input gate.

$$\mathrm{c}_t = \mathrm{f}_t \odot \mathrm{c}_{t-1} + \mathrm{i}_t \odot \tilde{\mathrm{c}}_t, \tag{4}$$

where $\odot$ denotes element-wise multiplication.
Output Gate: The output gate generates the hidden state h_t based on the updated cell state.

$$\mathrm{o}_t = \sigma\left(\mathrm{W}_o \cdot [\mathrm{h}_{t-1}, \mathrm{x}_t] + \mathrm{b}_o\right), \tag{5}$$

$$\mathrm{h}_t = \mathrm{o}_t \odot \tanh\left(\mathrm{c}_t\right). \tag{6}$$

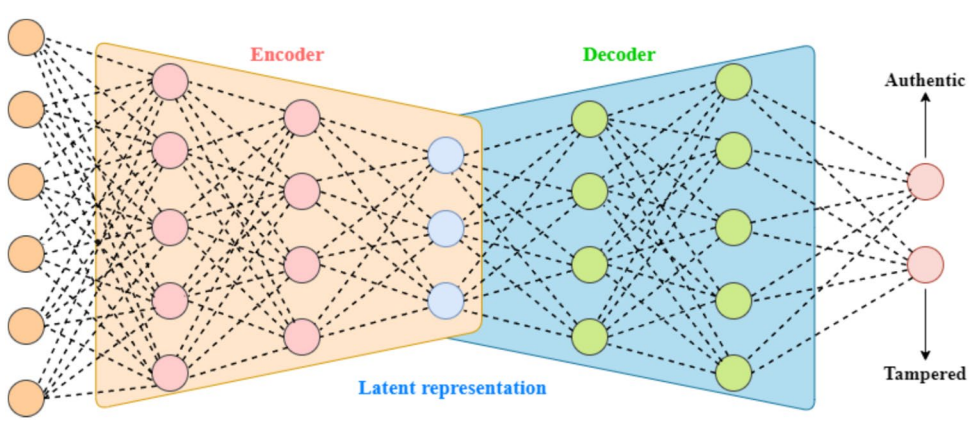

Fig. 12 Encoder-decoder architecture for ITD, showing feature extraction, bottleneck representation, and tampering localization. This illustration was reproduced from Neutelings (2021)

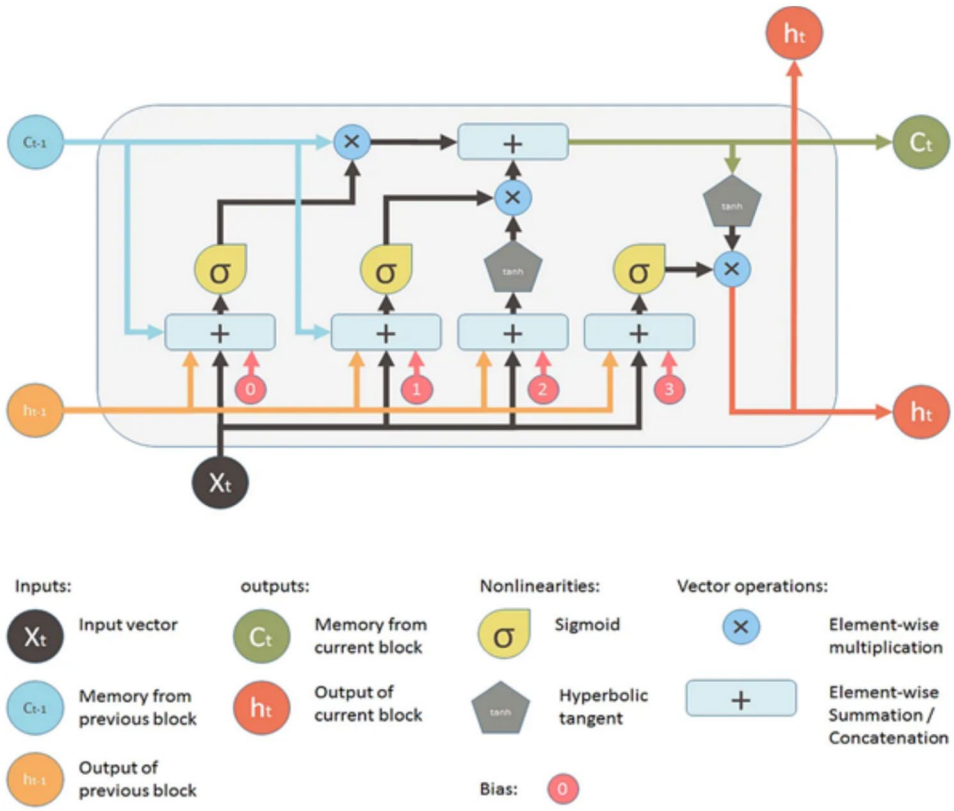

Fig. 13 Long short-term memory (LSTM) architecture adapted for ITD, capturing sequential dependencies for temporal forgery detection. This illustration was recreated from https://blog.mlreview.com/understanding-lstm-and-its-diagrams-37e2f46f1714

Table 3 Characteristics of widely used ITD datasets, including dataset size, tampering types, and ground-truth annotations

Datasets	No. of authentic images	No. of spliced images	No. of copy-move images	Numerous altered areas	Dimensions of tampered areas
CASIA V1.0 (Dong et al. 2013)	800	470	451	No	Small, Medium, Large
CASIA V2.0 (Dong et al. 2013)	7491	1849	3274	No	Small, Medium, Large
CoMoFoD (Tralic et al. 2013)	260	0	13520	Yes	Small
COVERAGE (Asudeh et al. 2019)	100	0	100	No	Small, Medium
MICC-F600 (Amerini et al. 2011)	440	0	160	Yes	Small, Medium, Large
MICC-F220 (Amerini et al. 2011)	110	0	110	Yes	Small, Medium, Large
COLUMBIA (Ng et al. 2004)	933	912	0	Yes	Small, Medium, Large

5 Image tampering datasets

In this section, we discuss widely used datasets for benchmarking ITD methods, such as CASIA, Columbia, and CoMoFoD. It highlights their characteristics, tampering scenarios, and utility in evaluating algorithmic performance. Detailed information on the image tampering datasets reviewed is provided in Table 3. This particular segment of the discourse intends to furnish a brief but thorough summary of the various datasets that have garnered popularity and widespread utilization in the realm of solving problems associated with Image Tampering Detection (ITD). It is a common occurrence that researchers undertaking investigations in this field tend to restrict their experimental procedures to a select few among the multitude of publicly accessible datasets, rather than engaging with the entirety of the available resources. A thorough and meticulous compilation of the pertinent information relating to the image tampering datasets that have been systematically reviewed is systematically presented in a tabular format.

1. *COLUMBIA DATASETS*: The first dataset to include spliced images was the Columbia Image Splicing Detection Evaluation Dataset (Ng et al. 2004; Pham and Park 2023), which was released in 2004. All 912 spliced and 933 legitimate photos in this dataset had basic splicing actions. Only 183 real and 180 spliced color photos made up the Columbia Uncompressed dataset in the second iteration. The resulting picture is approximately five to six times larger in this format. The photos in this

dataset were altered using straightforward alteration techniques that made it easy for human eyes to spot the tampering.

2. *CASIA*: The CASIA (Dong et al. 2013) Image Tampering Datasets are prominent benchmarks in digital image forensics, aiding the development and evaluation of tampering detection algorithms. CASIA v1.0, released in 2010, contains 800 images (400 original and 400 tampered) featuring basic tampering techniques like copy-move and splicing. Its relatively small size limits its use in training large models. CASIA v2.0, released in 2011, significantly expanded the dataset to 12,614 images (7,491 original and 5,123 tampered), incorporating more diverse tampering scenarios such as post-processing effects like blurring and compression, making it suitable for machine and deep learning methods. Both datasets extensively evaluate tampering detection, focusing on copy-move and splicing techniques. They provide controlled, high-quality data and remain critical benchmarks for comparing forensic algorithms.
3. *FORENSICS DATASET*: In 2013, the Information Forensics and Security Technical Committee (IFSTC) published the forensic dataset (image forensics challenge 2013) as part of the inaugural IEEE Forensics challenge. This dataset has 700 testing pictures and 450 training photos, each with dimensions of 2018 x 1536 pixels.
4. *CoMoFoD DATASET*: The CoMoFoD (Copy-Move Forgery Detection) dataset (Tralic et al. 2013) serves as a thorough benchmark for assessing algorithms aimed at identifying copy-move forgeries in digital images. It comprises two subsets: a minor set consisting of 200 forged and 200 legitimate photographs, and a major set including 25,000 forged images. The dataset encompasses several copy-move forgery scenarios, including scaling, rotation, JPEG compression, and illumination variations, rendering it appropriate for evaluating the resilience of detection methods. All forgeries in the dataset are marked with ground truth masks, facilitating accurate performance assessment. CoMoFoD is extensively employed in research because of its variety and capacity to simulate authentic tampering scenarios.
5. *COVERAGE DATASET*: This dataset (Asudeh et al. 2019) is a leading image forensics tool for detecting copy-move forgeries. This technique is used to modify high-resolution photos under different situations to test detection systems. There are 200 photos in the collection, 100 original and 100 manipulated. Focusing on scale, rotation, and noise addition allows thorough algorithmic performance assessment. Academic researchers utilize the dataset to benchmark traditional and DL-based forgery detection methods since it provides annotated ground realities for exact evaluation.
6. *MICC-F600 DATASET*: The MICC-F600 dataset (Amerini et al. 2011) serves as a benchmark specifically intended for the assessment of CMFD techniques. The collection comprises 600 high-quality pictures, consisting of 440 real pictures and 160 photoshopped pictures. The manipulated photographs are created by realistic CMF techniques, frequently incorporating diverse post-processing alterations such as blurring, rotation, and compression to emulate real-world difficulties.
7. *MICC-F220 DATASET*: The MICC-F220 dataset (Amerini et al. 2011) is a standard dataset created for assessing copy-move forgery detection (CMFD) algorithms. It comprises 220 high-resolution photos, categorized into 110 genuine images and 110 altered ones. The altered photographs are produced by copy-move forgeries employing realistic post-processing methods, including rotation, scaling, and compression, to accurately simulate real-world manipulation situations.

Table 3 compares widely used datasets in image tampering detection. CASIA v1.0 and v2.0 remain popular benchmarks, though v1.0 is too small for training deep learning models. CoMoFoD provides large-scale copy-move scenarios with scaling and illumination variations, making it valuable for testing robustness. MICC-F220 and MICC-F600 emphasize realistic copy-move forgeries, while COVERAGE focuses on geometric transformations. Columbia datasets, though smaller, are important for early splicing research. These differences highlight the need for dataset diversity when evaluating ITD models.

6 Evaluation metrics

The performance of image tampering detection (ITD) methods is typically evaluated using quantitative metrics that assess both classification accuracy and localization ability. Selecting appropriate metrics is essential, as different tampering scenarios (e.g., copy-move vs. splicing) demand emphasis on different aspects of performance. This section discusses the most widely adopted metrics, including accuracy, precision, recall, F_1-score, Intersection over Union (IoU), and Area Under the Curve (AUC), and highlights their significance in ITD research (Goodfellow et al. 2016; Pham and Park 2023; Wu et al. 2018; Pham et al. 2020; Jaccard 1912; Powers 2020).

6.1 Pixel-level metrics

Localization-based approaches, pixel-level measures are widely used to quantify the overlap between predicted tampered regions and ground-truth masks. Precision (M_P) measures the fraction of correctly identified tampered pixels among all pixels predicted as tampered (Powers 2020).

$$M_P = \frac{\#\text{ correctly identified pixels}}{\#\text{ all identified pixels}}. \tag{7}$$

Recall (M_R) captures the ability of the detector to identify all tampered pixels (Powers 2020).

$$M_R = \frac{\#\text{ correctly identified pixels}}{\#\text{ all spliced pixels}}. \tag{8}$$

Since precision and recall are often in tension, where high precision may come at the cost of low recall and vice versa, their harmonic mean, the F_1-score, provides a balanced measure (Chinchor and Sundheim 1993).

$$M_F = 2 \cdot \frac{M_P \cdot M_R}{M_P + M_R}. \tag{9}$$

The F_1-score is particularly valuable when tampered regions are small relative to the overall image, as it balances false positives and false negatives.

6.2 Image-level metrics

Classification-based ITD methods, accuracy remains a straightforward and commonly reported metric (Pham and Park 2023).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \tag{1}$$

where *TP*, *TN*, *FP*, and *FN* represent true positives, true negatives, false positives, and false negatives, respectively. While accuracy provides a general overview, it can be misleading when datasets are imbalanced, for instance, when authentic images significantly outnumber tampered ones (Powers 2020). In such cases, a model may achieve high accuracy by trivially predicting most samples as authentic.

6.3 Advanced metrics

To capture finer aspects of performance, several advanced metrics are also used in ITD research.

- *Intersection over Union (IoU):* Quantifies the overlap between predicted and ground-truth tampered regions, providing a stricter measure of localization accuracy than recall or precision alone (Jaccard 1912; Long et al. 2015).
- *Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC):* Measure the trade-off between true positive and false positive rates across thresholds. AUC is widely used in evaluating binary classifiers and is especially valuable for comparing models across datasets with varying thresholds (Fawcett 2006).
- *Equal Error Rate (EER):* Used in biometric and forensic contexts, EER identifies the point where false acceptance and false rejection rates are equal, offering an intuitive indicator of verification reliability (Wayman 2001).

Each metric provides complementary insights into model performance. Pixel-level metrics (precision, recall, F_1, IoU) are best suited for evaluating localization-based methods such as copy-move and splicing detection, while image-level metrics (accuracy, AUC, EER) are more appropriate for classification-based approaches, including tampering presence detection (Wu et al. 2018; Pham et al. 2020). Importantly, relying on a single metric can obscure weaknesses; for example, a high accuracy may hide poor localization capability. Consequently, recent ITD studies increasingly report multiple metrics to provide a comprehensive evaluation of detection and localization performance.

As shown in Table 4, no single metric is sufficient to fully capture the performance of ITD methods. For instance, a model may achieve high accuracy in classifying authentic versus tampered images while failing to localize manipulated regions effectively. Similarly, precision emphasizes minimizing false alarms, whereas recall prioritizes capturing all tampered pixels. Therefore, reporting multiple complementary metrics such as accuracy, precision, recall, F_1-score, and IoU, provides a more balanced and transparent assessment of both classification and localization performance. This multi-metric evaluation is particularly critical for ITD research, where datasets are often imbalanced and manipulation types vary widely in scale, subtlety, and context.

7 Conclusion & future scope

This study presented a comprehensive analysis of deep learning techniques for image tampering detection (ITD), underscoring their superior performance over conventional methods in terms of accuracy, robustness, and feature representation. A wide range of tampering detection methodologies, including copy-move, splicing, resampling,

Table 4 Summary of evaluation metrics for image tampering detection (ITD), including their definitions, best-suited applications, and key limitations. These metrics, ranging from accuracy, precision, recall, and F_1-score to IoU, AUC, and EER, provide complementary insights into both classification and localization performance

Metric	Definition	Evaluation focus	Limitations
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$; overall proportion of correct predictions (Pham and Park 2023; Powers 2020)	Image-level classification (authentic vs tampered)	Misleading in imbalanced datasets; ignores localization quality
Precision (M_P)	Fraction of correctly identified tampered pixels among predicted tampered pixels (Powers 2020)	Pixel-level localization when false positives must be minimized	May be high even if many tampered pixels are missed (low recall)
Recall (M_R)	Fraction of correctly identified tampered pixels among all ground-truth tampered pixels (Powers 2020)	Pixel-level localization when minimizing missed detections is critical	High recall may come at the cost of many false positives
F_1-score (M_F)	Harmonic mean of precision and recall (Chinchor and Sundheim 1993)	Balanced evaluation of pixel-level localization	May underrepresent performance when one of the precision/recall is extreme
IoU	Ratio of overlap to union between predicted and ground-truth tampered regions (Jaccard 1912; Long et al. 2015)	Strict evaluation of localization accuracy	Sensitive to boundary errors; requires accurate ground-truth masks
AUC (ROC)	Area under the ROC curve; measures trade-off between TPR and FPR across thresholds (Fawcett 2006)	Image-level classification across multiple thresholds	Less informative for localization tasks
EER	Error rate where false acceptance = false rejection (Wayman 2001)	Forensic and authentication tasks	Not widely reported; threshold-dependent

and statistical analysis, were examined, with clear distinctions drawn between active and passive authentication approaches. Prominent architectures such as CNNs, encoder-decoder models, and RNNs were reviewed, highlighting their contributions to tampering localization and classification. Benchmark datasets and evaluation metrics were also surveyed, emphasizing their role in advancing ITD research and applications across domains such as forensics, journalism, and social media. Despite these advances, challenges remain in achieving cross-dataset generalization, improving computational efficiency for real-time use, and defending against increasingly sophisticated tampering techniques. Looking ahead, transformer-based architectures offer promise for capturing long-range dependencies, while self-supervised and few-shot learning approaches may reduce reliance on large annotated datasets. The integration of explainable AI (XAI) is crucial for enhancing transparency and trust in forensic investigations, and the design of lightweight, energy-efficient models will support deployment on edge devices such as smartphones and surveillance systems. Furthermore, multimodal strategies that combine image, video, and metadata analysis represent a promising direction for countering complex manipulations. Progress in these areas, alongside the development of more diverse benchmark datasets and cross-disciplinary collaboration, will be essential for advancing ITD and strengthening digital trust in an evolving technological landscape.

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Data availability Data sharing is not applicable to this article as no new data were created or analyzed in this study. The datasets reviewed in this study (including CASIA, Columbia, and CoMoFoD) are publicly available and cited within the references.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals.

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